(iii) If the decimal is shifted to the left, the exponent n is positive. If the decimal is shifted to the right, the exponent n is negative.

Expressing a number N in the form of $N = a \times 10^n$ where, $1 \leq a < 10$ and 'n' is an integer is called as Scientific Notation.

The following table of base 10 examples may make things clearer:

Decimal notation	Scientific notation
100	1×10^2
1,000	1×10^3
10,000	1×10^4
1,00,000	1×10^5
10,00,000	1×10^6
1,00,00,000	1×10^7

Decimal notation	Scientific notation
0.01	1×10^{-2}
0.001	1×10^{-3}
0.0001	1×10^{-4}
0.00001	1×10^{-5}
0.000001	1×10^{-6}
0.0000001	1×10^{-7}

Let us look into few more examples.

Example 2.28 Express in scientific notation (i) 9768854 (ii) 0.04567891 (iii) 72006865.48

Solution

(i) $9\ 7\ 6\ 8\ 8\ 5\ 4\ .\ 0 = 9.768854 \times 10^6$

6 5 4 3 2 1

The decimal point is to be moved six places to the left. Therefore $n = 6$.

(ii) $0\ .\ 0\ 4\ 5\ 6\ 7\ 8\ 9\ 1 = 4.567891 \times 10^{-2}$

1 2

The decimal point is to be moved two places to the right. Therefore $n = -2$.

(iii) $7\ 2\ 0\ 0\ 6\ 8\ 6\ 5\ .\ 4\ 8 = 7.200686548 \times 10^7$

7 6 5 4 3 2 1

The decimal point is to be moved seven places to the left. Therefore $n = 7$.

2.8.2 Converting Scientific Notation to Decimal Form

The reverse process of converting a number in scientific notation to the decimal form is easily done when the following steps are followed:

(i) Write the decimal number.

2-Real Numbers.indd 71 02/12/2022 21:34:12